

Vasicek Companion Project

Research Question

To what extent does the numerical solution of the PDE associated with the Vasicek model faithfully reproduce the model's analytical properties, and what are the practical limits of this approach for calibrating a real-world yield curve?

Bourhan Kayal

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1 Introduction

1.1 Interest-rate context

Interest rates play a central role in modern finance, especially for valuing bonds, interest-rate derivatives, and managing interest-rate risk. The term structure of interest rates, which describes the relationship between maturity and the level of rates, is a fundamental object in financial markets.

To model the term structure in a coherent way, it is common to introduce an instantaneous interest rate, called the *short rate*, which serves as the state variable of the model. The dynamics of the short rate then allow one to derive zero-coupon bond prices and, by extension, the full set of spot and forward rates observed in the market.

This project fits within this framework and aims to study a classic short-rate model: the Vasicek model, with a focus on its mathematical formulation, its translation into a partial differential equation (PDE), and the numerical methods used to solve this PDE.

Project objective and guiding thread.

The goal of this project is not only to present the Vasicek model, but to assess to what extent its numerical PDE-based solution faithfully reproduces its analytical properties, and to identify the practical limitations of this approach when calibrating the model to a real yield curve.

This guiding thread will successively structure the theoretical study of the model, the implementation of numerical methods, and the analysis of calibration results and their limitations.

1.2 Presentation of the Vasicek model

The Vasicek model is a short-rate model introduced in 1977, widely used in quantitative finance because of its analytical simplicity and pedagogical value. It assumes that the short rate r_t follows a mean-reverting stochastic dynamics.

Under the risk-neutral measure, the short rate is modeled by the following stochastic differential equation:

$$dr_t = a(b - r_t) dt + \sigma dW_t,$$

where $a > 0$ is the speed of mean reversion toward the long-term level b , $\sigma > 0$ is the rate volatility, and W_t is a standard Brownian motion.

This model captures two essential behaviors observed in interest rates:

- a long-term tendency toward an equilibrium level b ,
- random fluctuations around this level due to market shocks.

The Vasicek equation admits an explicit solution, which makes it possible to directly characterize the expectation and variance of the short rate. One can show that the rate evolves on average toward a long-term level and that its variance remains bounded. Since the solution is Gaussian, the model allows negative rates, which is one of its main limitations.

2 Theory

2.1 Short-rate dynamics

The Vasicek equation is a linear stochastic differential equation that admits an explicit solution. Using the integrating factor e^{at} , one obtains:

$$r_t = r_0 e^{-at} + b(1 - e^{-at}) + \sigma \int_0^t e^{-a(t-s)} dW_s.$$

This expression fully characterizes the process. In particular, the conditional expectation and variance of the short rate are:

$$\mathbb{E}[r_t] = b + (r_0 - b)e^{-at},$$

$$\text{Var}(r_t) = \frac{\sigma^2}{2a} (1 - e^{-2at}).$$

As $t \rightarrow \infty$, the process admits a stationary distribution with mean b and variance $\sigma^2/(2a)$. The parameter a therefore controls both the speed of mean reversion and the long-term amplitude of fluctuations.

2.2 From the SDE to the PDE

Consider a financial product with value $V(t, r)$, depending on time and the current short rate. The short rate r_t is assumed to follow Vasicek dynamics under the risk-neutral measure. Applying Itô's formula to $V(t, r_t)$ yields:

$$dV(t, r_t) = \left(\frac{\partial V}{\partial t} + a(b - r_t) \frac{\partial V}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial r^2} \right) dt + \sigma \frac{\partial V}{\partial r} dW_t.$$

Introduce the infinitesimal generator $\mathcal{L}$ associated with the short-rate process r_t , defined by:

$$\mathcal{L}V = a(b - r) \frac{\partial V}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial r^2}.$$

In finance, the economically relevant quantity is the *discounted* price process. Under no-arbitrage, one imposes that

$$e^{-\int_0^t r_s ds} V(t, r_t)$$

is a martingale under the risk-neutral measure.

The full proof of this property, which relies on applying Itô's formula to the discounted process and canceling the drift term, is not detailed here. *We assume* that this martingale condition leads to the Vasicek pricing PDE:

$$\frac{\partial V}{\partial t} + \mathcal{L}V - rV = 0.$$

Probabilistic interpretation: Feynman–Kac formula (assumed)

The pricing PDE fits into the general framework of the Feynman–Kac formula, which provides a fundamental link between solving a parabolic PDE and computing a conditional expectation under the risk-neutral measure.

Under sufficient regularity assumptions on the coefficients, the solution of

$$\partial_t V(t, r) + \mathcal{L}V(t, r) - rV(t, r) = 0, \quad V(T, r) = \Phi(r),$$

admits the probabilistic representation

$$V(t, r) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r_s ds \right) \Phi(r_T) \middle| r_t = r \right].$$

For a zero-coupon bond, $\Phi(r) \equiv 1$, and one recovers the discounted expectation:

$$P(t, r) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r_s ds \right) \middle| r_t = r \right].$$

This probabilistic viewpoint justifies the PDE-based numerical approach: it is a deterministic method that is theoretically equivalent to a Monte Carlo valuation based on the same short-rate dynamics.

2.3 Characterization of the PDE

The PDE associated with the Vasicek model can be written (notation consistent with the report) as:

$$\frac{\partial V}{\partial t} + a(b - r) \frac{\partial V}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial r^2} - rV = 0.$$

A general linear second-order PDE in variables (r, t) can be written as:

$$A \frac{\partial^2 V}{\partial r^2} + B \frac{\partial^2 V}{\partial r \partial t} + C \frac{\partial^2 V}{\partial t^2} + D \frac{\partial V}{\partial r} + E \frac{\partial V}{\partial t} + FV = 0,$$

with discriminant:

$$\Delta = B^2 - 4AC.$$

The zero-coupon PDE can be written in the standard evolution form:

$$\partial_t P(t, r) = -a(b - r) \partial_r P(t, r) - \frac{1}{2} \sigma^2 \partial_{rr} P(t, r) + r P(t, r).$$

It is therefore of *parabolic* type (second-order diffusion in r and first-order derivative in t), which motivates the use of implicit schemes (implicit Euler, Crank–Nicolson).

The parabolic nature of the PDE provides intuition about the solutions (diffusion/smoothing effect), which clarifies the relevance of implicit and Crank–Nicolson schemes used for numerical resolution.

2.4 Special case: zero-coupon bond

A zero-coupon bond with maturity T pays one unit of currency at time T , regardless of the evolution of the short rate. Its price at time t is denoted $P(t, T)$, or $P(t, r)$ when the dependence on r is made explicit.

The terminal condition is given by the payoff:

$$P(T, r) = 1 \quad \text{for all } r.$$

Under the risk-neutral measure, the zero-coupon price can also be written as a discounted expectation:

$$P(t, r) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r_s ds \right) \middle| r_t = r \right].$$

Applying the pricing PDE, the zero-coupon price satisfies:

$$\frac{\partial P}{\partial t} + a(b - r) \frac{\partial P}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial r^2} - rP = 0, \quad P(T, r) = 1.$$

This equation is the central object of the remainder of the project, both for analytical resolution and for numerical study.

2.5 Analytical solution for the zero-coupon bond

In the Vasicek model, one can obtain a closed-form expression for the zero-coupon price. Define the time to maturity

$$\tau = T - t, \quad Q(\tau, r) = P(T - \tau, r).$$

Using $\partial_t P = -\partial_\tau Q$, the zero-coupon PDE becomes the forward PDE

$$\partial_\tau Q = a(b - r) \frac{\partial Q}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 Q}{\partial r^2} - rQ, \quad Q(0, r) = 1.$$

Seek an exponentially affine solution

$$Q(\tau, r) = A(\tau) e^{-B(\tau)r}.$$

Substituting into the PDE and matching terms in r yields the ODE system:

$$B'(\tau) = 1 - aB(\tau), \quad B(0) = 0, \quad \frac{A'(\tau)}{A(\tau)} = -abB(\tau) + \frac{1}{2}\sigma^2 B(\tau)^2, \quad A(0) = 1.$$

The first equation gives

$$B(\tau) = \frac{1 - e^{-a\tau}}{a}.$$

Then $A(\tau)$ is obtained in closed form:

$$A(\tau) = \exp \left[\left(b - \frac{\sigma^2}{2a^2} \right) (B(\tau) - \tau) - \frac{\sigma^2}{4a} B(\tau)^2 \right].$$

Finally,

$$P(t, T) = A(T - t) \exp(-B(T - t)r_t).$$

2.6 Spot and forward rates

The spot (zero-coupon) rate for maturity T is defined by:

$$R(t, T) = -\frac{1}{T-t} \ln P(t, T).$$

Using $P(t, T) = A(\tau)e^{-B(\tau)r_t}$ with $\tau = T - t$,

$$R(t, T) = -\frac{\ln A(\tau)}{\tau} + \frac{B(\tau)}{\tau} r_t.$$

In particular, for very long maturities ($\tau \rightarrow \infty$), $B(\tau) \rightarrow 1/a$ and thus $B(\tau)/\tau \rightarrow 0$, meaning the influence of the current short rate on very long rates vanishes.

The instantaneous forward rate is defined by:

$$f(t, T) = -\frac{\partial}{\partial T} \ln P(t, T).$$

Since $\tau = T - t$, one has $\partial_T = \partial_\tau$, hence:

$$f(t, T) = -\frac{A'(\tau)}{A(\tau)} + B'(\tau) r_t, \quad B'(\tau) = 1 - aB(\tau) = e^{-a\tau}.$$

2.7 Qualitative analysis of parameters

The closed-form zero-coupon price allows a qualitative analysis of how Vasicek parameters affect the term structure.

The parameter a , the mean-reversion speed, controls shock persistence. A larger a implies faster reversion to equilibrium, reduces the influence of the current short rate on long maturities, and tends to stabilize the yield curve.

The parameter b is the long-term level of the short rate. As maturity increases, the influence of the current rate fades and long rates are primarily driven by this equilibrium level. Increasing b therefore raises long-term rates.

Finally, the volatility σ appears only in the $A(\tau)$ factor. Increasing σ increases uncertainty about future rates, which lowers bond prices and increases the corresponding yields.

2.8 Consistency checks for the analytical solution

The analytical solution for the Vasicek zero-coupon bond can be checked through standard consistency tests. Let $\tau = T - t$.

Short-maturity limit

As $\tau \rightarrow 0$, the zero-coupon price must converge to its maturity payoff, i.e. 1. Indeed, $A(0) = 1$ and $B(0) = 0$, so:

$$P(T, T) = 1.$$

Moreover, for small maturities, $B(\tau) \sim \tau$, and:

$$P(t, T) \approx e^{-r_t \tau}.$$

Hence:

$$R(t, T) = -\frac{1}{\tau} \ln P(t, T) \xrightarrow{\tau \rightarrow 0} r_t.$$

This is economically consistent: the very short-term zero-coupon rate matches the instantaneous short rate.

Long-maturity limit

As $\tau \rightarrow \infty$,

$$B(\tau) \longrightarrow \frac{1}{a}, \quad \frac{B(\tau)}{\tau} \longrightarrow 0.$$

In

$$R(t, T) = -\frac{\ln A(\tau)}{\tau} + \frac{B(\tau)}{\tau} r_t,$$

the coefficient multiplying r_t tends to zero. Thus, for long maturities, the effect of the current short rate vanishes, and long-term rates are mainly governed by structural parameters, especially b . This matches the economic intuition that long rates reflect long-run expectations more than current market conditions.

Deterministic case $\sigma = 0$

When volatility is zero, the short rate is deterministic:

$$dr_t = a(b - r_t) dt,$$

and the zero-coupon price must satisfy:

$$P(t, T) = \exp \left(- \int_t^T r_s ds \right).$$

In this case,

$$A(\tau) = \exp(b(B(\tau) - \tau)),$$

so

$$P(t, T) = \exp(-b\tau - (r_t - b)B(\tau)).$$

Since

$$r_{t+s} = b + (r_t - b)e^{-as},$$

we have

$$\int_t^T r_s ds = b\tau + (r_t - b)B(\tau).$$

Both expressions coincide, confirming consistency in the noise-free case.

These checks confirm that the closed-form Vasicek solution is consistent both mathematically and economically. It provides a reliable reference for qualitative analysis and for validating the numerical methods studied next.

3 Numerical methods

The analytical resolution of the Vasicek pricing PDE is a valuable benchmark, but it is only possible due to the linearity and special structure of the model. In more general settings, or when assessing numerical robustness, it is necessary to rely on numerical methods to solve the pricing PDE.

This section presents the discretization choices, the construction of the discrete operator, and the time-stepping schemes used to approximate the zero-coupon solution.

3.1 Change of variable (time τ) and forward PDE

To work with an initial condition instead of a terminal condition, define:

$$\tau = T - t, \quad Q(\tau, r) = P(T - \tau, r).$$

The terminal condition $P(T, r) = 1$ becomes:

$$Q(0, r) = 1.$$

Using $\partial_t P = -\partial_\tau Q$, the zero-coupon PDE becomes:

$$\partial_\tau Q = a(b - r) \frac{\partial Q}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 Q}{\partial r^2} - rQ, \quad Q(0, r) = 1.$$

3.2 Operator form and numerical interpretation

Define the spatial operator:

$$\mathcal{A}Q = a(b - r) \frac{\partial Q}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 Q}{\partial r^2} - rQ.$$

Then:

$$\partial_\tau Q = \mathcal{A}Q.$$

This highlights the drift–diffusion–reaction structure:

- a *transport* term $a(b - r) \partial_r Q$,
- a *diffusion* term $\frac{1}{2} \sigma^2 \partial_{rr} Q$,

-
- a *reaction* term $-rQ$ due to discounting.

3.3 Spatial discretization

The state variable r lives on $\mathbb{R}$, which is not compatible with direct numerical implementation. We therefore truncate the domain, using the fact that under Vasicek the short rate is Gaussian.

At maturity T ,

$$\mathbb{E}[r_T] = b + (r_0 - b)e^{-aT}, \quad \text{Var}(r_T) = \frac{\sigma^2}{2a}(1 - e^{-2aT}).$$

Choose a computational interval:

$$[r_{\min}, r_{\max}] = \left[\mathbb{E}[r_T] - k\sqrt{\text{Var}(r_T)}, \mathbb{E}[r_T] + k\sqrt{\text{Var}(r_T)} \right],$$

with $k > 0$ (typically $k = 4$ or $k = 5$) to capture most probability mass.

In the numerical experiments below, we use a fixed interval large enough to make truncation error negligible while keeping computation reasonable. A typical choice is:

$$r \in [-2\%, 8\%],$$

which covers plausible short-rate values (including slightly negative rates) and stabilizes the schemes without imposing artificial boundary values.

Discretize into M subintervals of size Δr :

$$r_i = r_{\min} + i\Delta r, \quad i = 0, \dots, M, \quad \Delta r = \frac{r_{\max} - r_{\min}}{M}.$$

Approximate spatial derivatives with centered finite differences (for $i = 1, \dots, M-1$):

$$\frac{\partial Q}{\partial r}(\tau, r_i) \approx \frac{Q_{i+1} - Q_{i-1}}{2\Delta r}, \quad \frac{\partial^2 Q}{\partial r^2}(\tau, r_i) \approx \frac{Q_{i+1} - 2Q_i + Q_{i-1}}{\Delta r^2}.$$

3.4 Time discretization and notation Q_i^n

Discretize $\tau \in [0, T]$ into N steps:

$$\tau_n = n\Delta\tau, \quad n = 0, \dots, N, \quad \Delta\tau = \frac{T}{N}.$$

Define:

$$Q_i^n \approx Q(\tau_n, r_i).$$

3.5 Boundary conditions

On $[r_{\min}, r_{\max}]$, impose homogeneous Neumann conditions:

$$\frac{\partial Q}{\partial r}(\tau, r_{\min}) = 0, \quad \frac{\partial Q}{\partial r}(\tau, r_{\max}) = 0.$$

Numerically:

$$Q_0^n = Q_1^n, \quad Q_M^n = Q_{M-1}^n, \quad \forall n.$$

Several strategies exist: incorporate them into the matrix (modify first/last rows) or enforce them after solving the interior system. In our implementation, we use a simple and robust projection: after each time step, project the solution onto the set of vectors satisfying

$$Q_0^n = Q_1^n, \quad Q_M^n = Q_{M-1}^n.$$

This is consistent with the discretization order and avoids special boundary coefficients, while allowing easy automated checks of Neumann conditions at each iteration.

Remark: truncation choice and boundary condition impact

Numerical resolution requires truncating the short-rate domain, theoretically $\mathbb{R}$. The choice of $[r_{\min}, r_{\max}]$ (via k standard deviations) aims to capture most of the probability mass of r_T under Vasicek.

Homogeneous Neumann conditions express that, for extreme rate levels, the sensitivity of the bond price to the short rate becomes negligible. This avoids imposing artificial values (Dirichlet conditions) that could create boundary bias.

In practice, there is a trade-off between domain width and computational cost: too

narrow introduces truncation error (bias), too wide increases system size unnecessarily. Numerical tests suggest that for usual k values, truncation error is negligible compared to discretization error.

3.6 Implicit Euler scheme: tridiagonal system and coefficients

Applying implicit Euler to $\partial_\tau Q = \mathcal{A}Q$ gives:

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta\tau} = \mathcal{A}_h[Q^{n+1}]_i, \quad i = 1, \dots, M-1,$$

where $\mathcal{A}_h$ is the discretized spatial operator. Expanding (centered differences) yields the tridiagonal system:

$$-\alpha_i Q_{i-1}^{n+1} + \beta_i Q_i^{n+1} - \gamma_i Q_{i+1}^{n+1} = Q_i^n, \quad i = 1, \dots, M-1,$$

with

$$\alpha_i = \Delta\tau \left(\frac{\sigma^2}{2(\Delta r)^2} - \frac{a(b-r_i)}{2\Delta r} \right), \quad \beta_i = 1 + \Delta\tau \left(r_i + \frac{\sigma^2}{(\Delta r)^2} \right), \quad \gamma_i = \Delta\tau \left(\frac{\sigma^2}{2(\Delta r)^2} + \frac{a(b-r_i)}{2\Delta r} \right).$$

Neumann conditions are enforced numerically:

$$Q_0^n = Q_1^n, \quad Q_M^n = Q_{M-1}^n, \quad \forall n.$$

In our implementation, these are not embedded directly into the matrix. After each time step solve, we project the solution by explicitly setting boundary values equal to their neighbors.

This approach is simple, stable, and equivalent (at the chosen discretization order) to an explicit boundary-equation modification.

This scheme is first-order in time and unconditionally stable, making it a robust choice for validating the spatial discretization and the boundary condition treatment.

3.7 Explicit Euler scheme and stability condition (CFL)

For reference and to illustrate stability constraints, we also consider explicit Euler:

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta\tau} = \mathcal{A}_h[Q^n]_i, \quad \text{so} \quad Q_i^{n+1} = Q_i^n + \Delta\tau \mathcal{A}_h[Q^n]_i, \quad i = 1, \dots, M-1.$$

This scheme is easy to implement but *conditionally stable*. For drift–diffusion–reaction PDEs, a CFL-type condition typically requires $\Delta\tau$ to be sufficiently small relative to Δr .

In practice, the diffusion term often provides the most restrictive constraint:

$$\Delta\tau \lesssim \frac{(\Delta r)^2}{\sigma^2},$$

(up to a constant), making explicit Euler costly for fine spatial grids. Therefore, explicit Euler is mainly used here as a *pedagogical tool* (instability demonstration) and as a comparison baseline, while implicit schemes (implicit Euler and Crank–Nicolson) are preferred for robust computations.

3.8 Crank–Nicolson scheme

To improve time accuracy, we use Crank–Nicolson:

$$\frac{Q^{n+1} - Q^n}{\Delta\tau} = \frac{1}{2} (\mathcal{A}_h Q^{n+1} + \mathcal{A}_h Q^n).$$

In matrix form, with L the tridiagonal matrix representing $\mathcal{A}_h$ on interior points:

$$\left(I - \frac{\Delta\tau}{2} L\right) Q^{n+1} = \left(I + \frac{\Delta\tau}{2} L\right) Q^n.$$

This scheme is second-order in time and offers strong stability for parabolic PDEs, with better accuracy than implicit Euler at comparable cost.

Remark: accuracy and robustness of Crank–Nicolson

Crank–Nicolson is second-order accurate in time and very precise, but it can generate numerical oscillations if the time step is too large or if transport (convection) dominates.

For the Vasicek zero-coupon bond, the terminal condition $P(T, r) = 1$ is smooth and the operator is well-behaved, which limits such oscillations. Still, this highlights the

importance of choosing Δt and the classical trade-off between accuracy (Crank–Nicolson) and robustness (implicit Euler, more dissipative), especially for less regular payoffs.

3.9 Solution strategy and complexity

At each time step, the linear system involves a sparse tridiagonal matrix. Two natural solver families are:

- a dedicated tridiagonal solver (Thomas algorithm) in $\mathcal{O}(M)$,
- a general sparse solver (sparse LU), where factorization is paid once and reused.

In our implementation, a sparse LU factorization (CSC format, via `splu`) of the scheme matrix (implicit Euler or Crank–Nicolson) is performed *once* and reused at each time iteration, which significantly reduces computation time.

4 Numerical analysis

4.1 Numerical consistency tests

Several simple tests verify the consistency of the numerical solutions:

- the terminal condition $P(T, r) = 1$ is satisfied numerically (equivalently $Q(0, r) = 1$);
- the zero-coupon price is strictly positive;
- the price is decreasing in the short rate r ;
- Neumann boundary conditions are satisfied at each time step.

Note that in the Vasicek model, the short rate can be negative. In that case, the discount factor can exceed one, so it is economically consistent to observe zero-coupon prices slightly above one for sufficiently negative rates. This is taken into account when interpreting numerical results.

4.2 Economic properties: positivity, monotonicity, and numerical checks

Beyond numerical stability, a finite-difference pricing scheme should preserve expected *economic* properties of a zero-coupon bond price.

Positivity. By the Feynman–Kac representation,

$$P(t, r) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r_s ds \right) \middle| r_t = r \right],$$

so $P(t, r) > 0$ for all (t, r) . Numerically, we verify at each time step that $Q(\tau, r) = P(T - \tau, r)$ remains strictly positive on the grid.

Bounds and interpretation under negative rates. Because Vasicek allows $r_t < 0$ with non-zero probability, the discount factor $\exp(-\int r_s ds)$ can exceed 1. It is therefore consistent to observe $P(t, r) > 1$ for sufficiently negative r . This is considered in the interpretation.

Monotonicity in the short rate. Economically, a higher initial short rate implies stronger discounting and thus a lower price: $r \mapsto P(t, r)$ should be decreasing. This is apparent from the analytical solution $P(t, T) = A(\tau)e^{-B(\tau)r}$ with $B(\tau) > 0$, and is checked numerically by verifying that $P(0, r_i)$ decreases with r_i .

Systematic numerical checks. The implementation enforces:

- **Initial condition:** $Q(0, r) = 1$ (equivalent to $P(T, r) = 1$).
- **Positivity:** $Q(\tau_n, r_i) > 0$ for all n, i .
- **Monotonicity:** $P(0, r_i)$ decreasing in r_i .
- **Neumann conditions:** $Q_0^n = Q_1^n$ and $Q_M^n = Q_{M-1}^n$.

These checks are not a formal proof of preservation, but they provide robust validation of solver consistency on the considered grids.

4.3 Python implementation: sparse matrices and forward stepping in τ

Spatial finite differences yield tridiagonal linear systems at each time step. The Python implementation uses sparse matrices (CSC format) via `scipy.sparse`. Systems are solved using sparse LU factorization (`splu`), performed *once* per scheme (implicit Euler or Crank–Nicolson) and reused at each time iteration. This is efficient because the system matrix does not depend on time n nor on the solution.

The computation proceeds forward in τ : the initial condition $Q(0, r) = 1$ is known, and we advance to $\tau = T$, producing $P(0, r) = Q(T, r)$. Numerically, at each step $n \rightarrow n + 1$, we solve:

$$A Q^{n+1} = \text{RHS}(Q^n),$$

where A is tridiagonal and RHS depends on the chosen scheme.

4.4 Stability and numerical properties of the schemes

Implicit Euler is *unconditionally stable* for parabolic PDEs, but first-order in time and introduces numerical diffusion (a more smoothed solution). Crank–Nicolson is also stable for parabolic PDEs and second-order in time. In return, Crank–Nicolson may show

oscillations when the time step is too large or dynamics are stiff (transport-dominated or strong diffusion), although this is limited here due to the smooth terminal condition of the zero-coupon bond.

Numerical results confirm the theoretical properties of Crank–Nicolson: second-order time accuracy, strong stability for the parabolic PDE, and no visible oscillations in the zero-coupon case.

A classic trade-off emerges: Crank–Nicolson provides higher time accuracy than implicit Euler at comparable computational cost, making it the preferred scheme for quantitative analysis.

4.5 Scheme comparison: error and convergence

The analytical closed-form solution provides a natural benchmark. We compare at initial time $t = 0$ (i.e. $\tau = T$) by computing:

$$e(r_i) = P_{\text{num}}(0, r_i) - P_{\text{ana}}(0, r_i),$$

and two global error norms:

$$\|e\|_{\infty} = \max_i |e(r_i)|, \quad \|e\|_2 = \left(\frac{1}{M+1} \sum_{i=0}^M e(r_i)^2 \right)^{1/2}.$$

Convergence studies are primarily reported in the L^{∞} norm, which highlights the maximum grid error. The L^2 norm leads to qualitatively similar conclusions.

Time convergence. Fix a sufficiently fine spatial grid (large M) to make spatial error negligible, then vary N . We observe numerically an order ≈ 1 for implicit Euler and ≈ 2 for Crank–Nicolson, consistent with theory.

Space convergence. Similarly, fix a fine time grid (large N) and vary M . Centered finite differences yield second-order convergence in space for both schemes (same spatial operator).

4.6 Probabilistic validation: PDE vs Monte Carlo (assumed)

Feynman–Kac provides:

$$P(0, r_0) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_0^T r_s ds \right) \right].$$

This enables independent validation by Monte Carlo simulation of the short rate under $\mathbb{Q}$.

Simulation scheme. Simulate Vasicek via Euler–Maruyama on $t_n = n\Delta t$:

$$r_{n+1} = r_n + a(b - r_n)\Delta t + \sigma\sqrt{\Delta t} Z_n, \quad Z_n \sim \mathcal{N}(0, 1).$$

Approximate the discount integral by a Riemann sum:

$$\int_0^T r_s ds \approx \sum_{n=0}^{N-1} r_n \Delta t.$$

For each path m ,

$$D^{(m)} = \exp \left(- \sum_{n=0}^{N-1} r_n^{(m)} \Delta t \right),$$

and estimate:

$$P_{\text{MC}}(0, r_0) = \frac{1}{M} \sum_{m=1}^M D^{(m)}.$$

Error measure and confidence interval. Statistical error decays as $1/\sqrt{M}$. We report the empirical standard deviation of $D^{(m)}$ to build a 95% confidence interval. We compare $P_{\text{MC}}(0, r_0)$ with the PDE price $P_{\text{PDE}}(0, r_0)$.

4.7 Computational cost and efficiency

The main cost comes from solving a linear system at each time step. Thanks to sparsity and tridiagonal structure, LU factorization is fast and reusable. In our implementation, the matrix is factorized once, and each time step reduces to forward/back substitution (very cheap). A runtime comparison shows both schemes have similar costs for a given N , with the main difference coming from the Crank–Nicolson right-hand side construction.

A simple benchmark in the notebook confirms that at comparable grids, implicit Euler

and Crank–Nicolson have similar runtimes; the main performance gain comes from the one-time sparse LU factorization reused at each step.

4.8 Numerical validation: summary of tests

Solutions are validated via automated checks:

- **Terminal/initial condition:** $P(T, r) = 1$ (equivalently $Q(0, r) = 1$).
- **Neumann conditions:** $Q_0^n = Q_1^n$ and $Q_M^n = Q_{M-1}^n$.
- **Positivity:** $P(t, r) > 0$.
- **Monotonicity:** $P(0, r)$ decreases with r .
- **Economic bounds:** for negative r , $P > 1$ can occur and is consistent in Vasicek.

These tests, together with the analytical benchmark, ensure the robustness of the numerical solver.

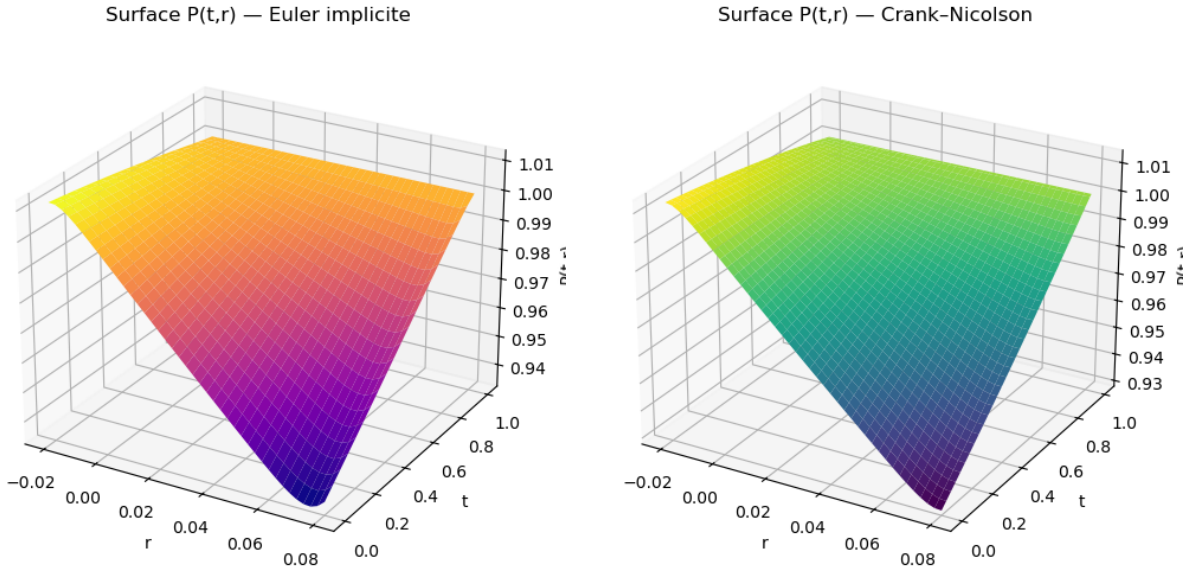
5 Results

5.1 3D visualization of the solution (surface $(\tau, r) \mapsto Q(\tau, r)$)

To visualize the full dynamics of the numerical solution, we plot the 3D surface

$$(\tau, r) \mapsto Q(\tau, r), \quad \tau \in [0, T], \quad r \in [r_{\min}, r_{\max}],$$

obtained with the different schemes. Recall that $P(t, r) = Q(T-t, r)$, so $P(0, r) = Q(T, r)$.



(a) Implicit Euler: surface $(\tau, r) \mapsto Q(\tau, r)$. (b) Crank–Nicolson: surface $(\tau, r) \mapsto Q(\tau, r)$.

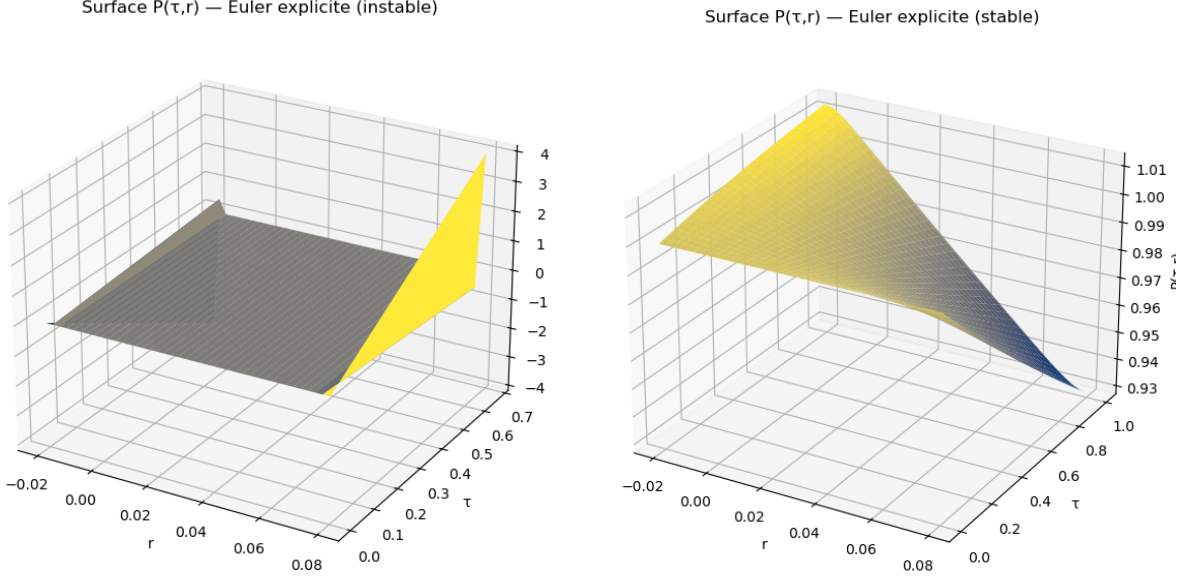
Figure 1: 3D surfaces of numerical solutions. Both methods produce a smooth and economically consistent surface (price decreasing in r and yielding $P(0, r) = Q(T, r)$).

5.2 Illustration: CFL condition and explicit Euler instability

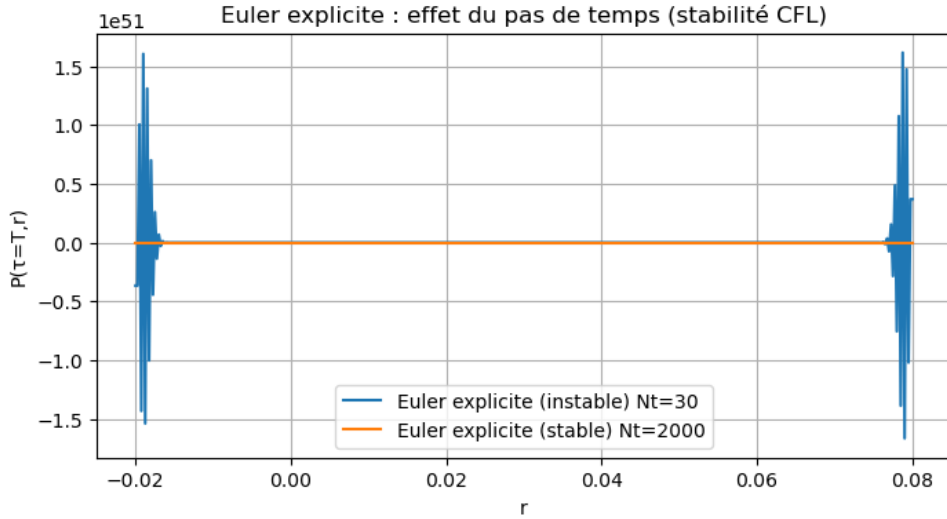
We illustrate the CFL-type stability constraint of explicit Euler. If $\Delta\tau$ is too large relative to Δr , the solution becomes unstable (oscillations, negative values, blow-up).

We compare two explicit Euler runs:

- $\Delta\tau$ too large: unstable scheme;
- smaller $\Delta\tau$: stable and qualitatively consistent solution.



(a) Explicit Euler with too large $\Delta\tau$: numerical instability. (b) Explicit Euler with smaller $\Delta\tau$: stable solution.



(c) Synthetic comparison of profiles ("dumbbell" visualization).

Figure 2: Numerical demonstration of the CFL constraint for explicit Euler: unstable surface (left), stable surface (right), and profile comparison.

5.3 Comparison of solutions

We compare the numerical PDE solution to the analytical closed-form solution for the zero-coupon bond price in Vasicek. The comparison is performed at $t = 0$, with the smooth terminal condition $P(T, r) = 1$.

Figure 3 shows the zero-coupon price $P(0, r)$ as a function of the initial short rate r , for the analytical solution and for the implicit Euler and Crank–Nicolson schemes. We

observe an almost perfect overlap between the analytical solution and Crank–Nicolson, while implicit Euler shows slight numerical diffusion, consistent with its first-order time accuracy.

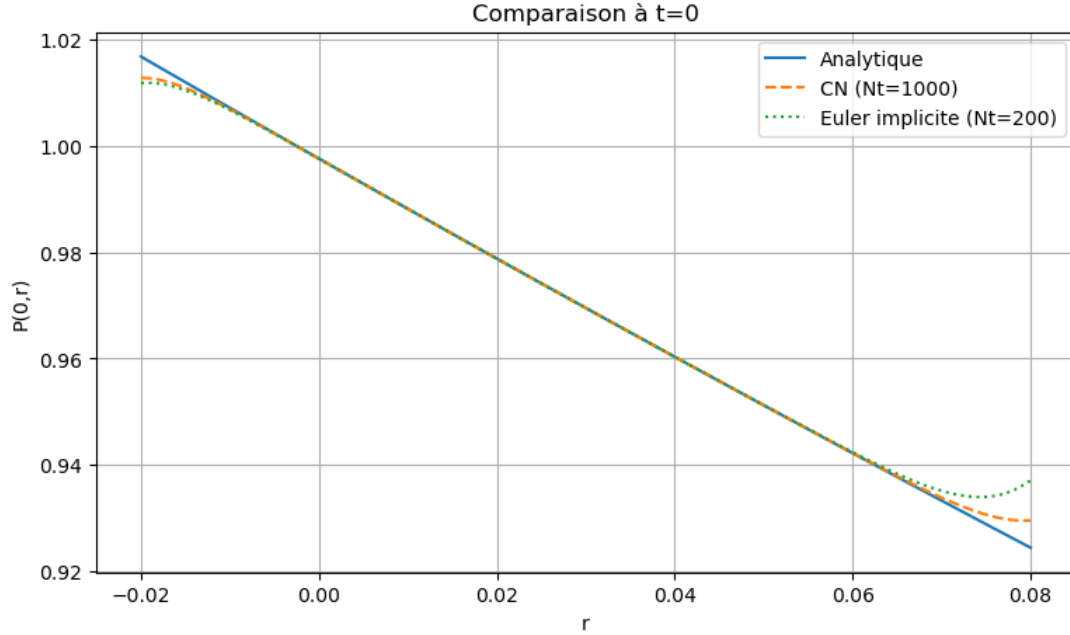


Figure 3: Zero-coupon price at $t = 0$: comparison between the analytical solution and the numerical schemes (Crank–Nicolson and implicit Euler).

Figure 4 displays the pointwise error

$$e(r) = P_{\text{num}}(0, r) - P_{\text{ana}}(0, r),$$

for both schemes. The error remains small and regular over the truncated domain, with no spurious oscillations, confirming scheme stability and the relevance of Neumann boundary conditions.

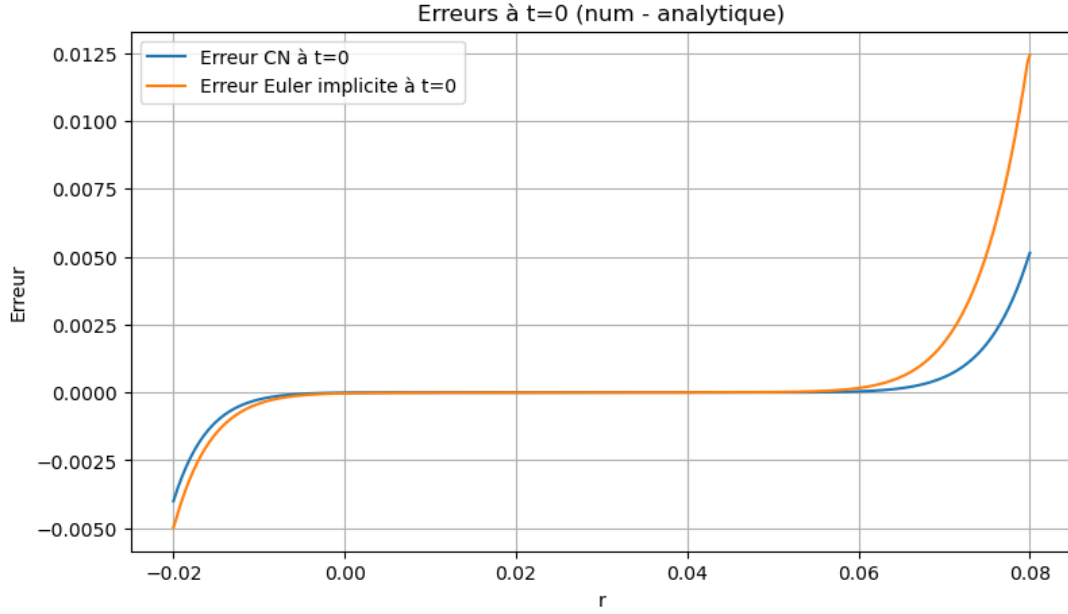


Figure 4: Error at $t = 0$ as a function of the initial short rate r for Crank–Nicolson and implicit Euler.

5.4 Time convergence analysis

To quantify time accuracy, we study convergence in time toward the analytical solution. Error is evaluated at $t = 0$ using the L^∞ norm:

$$\|e\|_\infty = \max_r |P_{\text{num}}(0, r) - P_{\text{ana}}(0, r)|.$$

Figure 5 shows this error as a function of Δt on a log scale, using a sufficiently fine spatial grid so that spatial error is negligible.

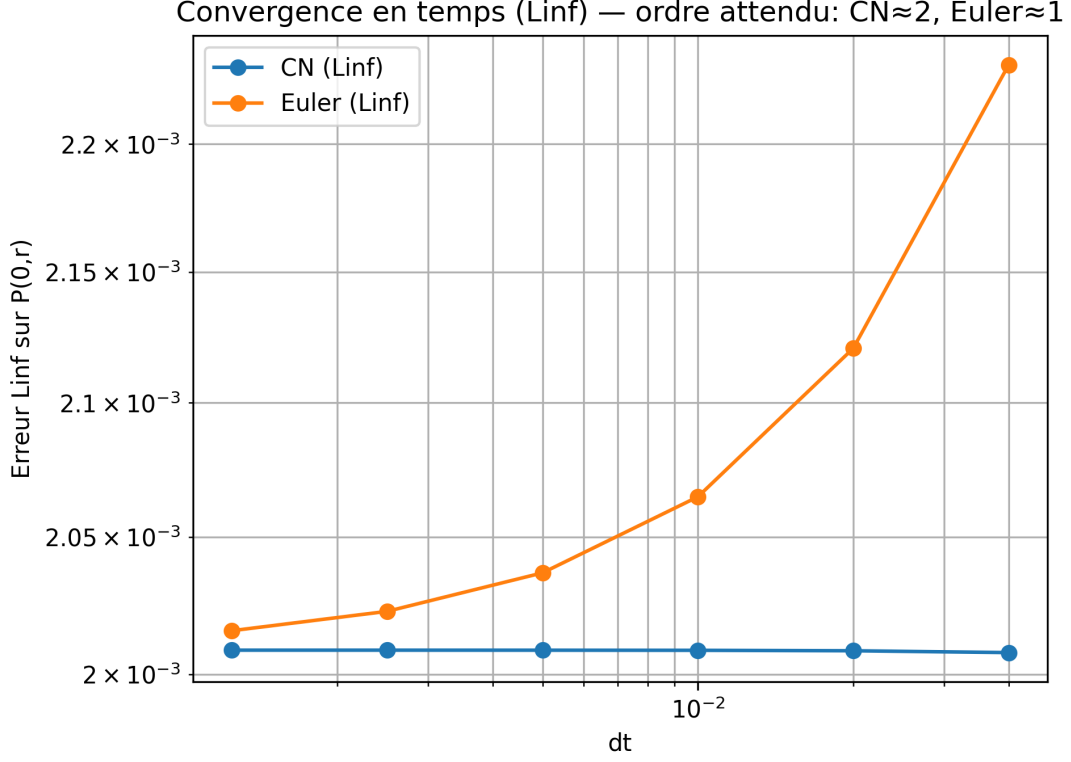


Figure 5: Time convergence (L^∞ norm) of Crank–Nicolson and implicit Euler.

We observe a convergence order close to 2 for Crank–Nicolson and close to 1 for implicit Euler, in agreement with theory.

5.5 Space convergence analysis

We now study spatial convergence by varying Δr , while choosing a sufficiently fine time step to make time error negligible. Error is again evaluated at $t = 0$ versus the analytical solution.

Figure 6 shows the L^∞ error versus Δr on a log scale for both schemes. Since both share the same centered spatial discretization, we expect the same space order.

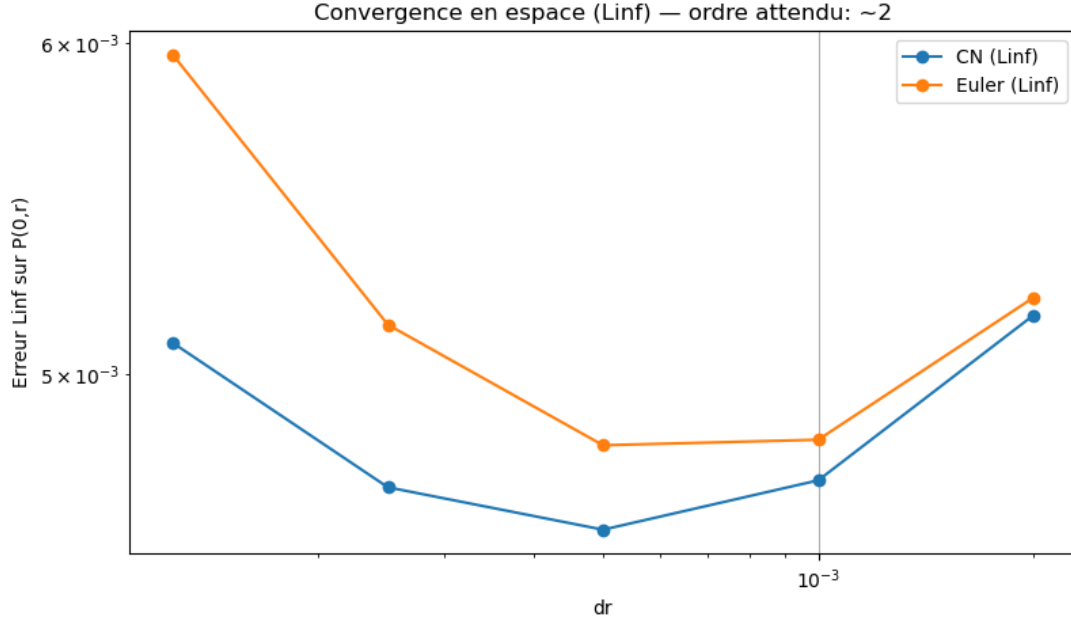


Figure 6: Space convergence (L^∞ norm) for Crank–Nicolson and implicit Euler.

We observe an order close to 2 for both schemes, consistent with second-order centered differences.

5.6 Cross-validation: PDE vs analytical vs Monte Carlo

The analytical solution is a strong benchmark, but it is also useful to validate the PDE solver via an independent method based on Feynman–Kac.

Recall:

$$P(0, r_0) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_0^T r_s ds \right) \right].$$

We estimate it via Monte Carlo simulation (Euler–Maruyama) and a Riemann-sum approximation for the integral. The Monte Carlo estimator naturally comes with a 95% confidence interval, allowing us to check whether the PDE value is statistically compatible.

Expected result. We expect:

$$P_{\text{PDE}}(0, r_0) \approx P_{\text{ana}}(0, r_0), \quad \text{and} \quad P_{\text{PDE}}(0, r_0) \in \text{IC}_{95\%}(P_{\text{MC}}(0, r_0)).$$

This is interesting because it compares:

- a *deterministic* method (PDE),

- a *stochastic* method (Monte Carlo),
- and a *closed-form* reference (analytical solution).

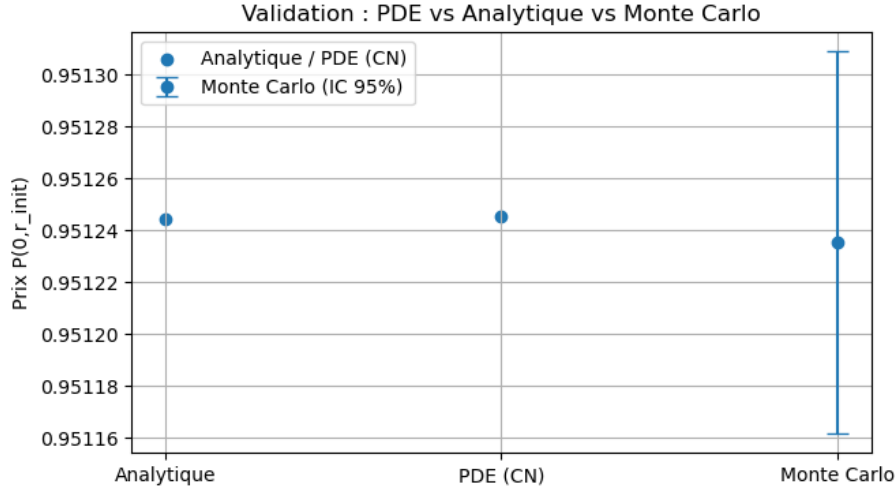


Figure 7: Feynman-Kac validation: comparison of $P(0, r_0)$ from the analytical solution, PDE solver, and Monte Carlo (95% CI).

Conclusion. If $P_{\text{PDE}}(0, r_0)$ lies within the Monte Carlo confidence interval, this validates the PDE resolution independently of the analytical solution and confirms overall implementation consistency.

5.7 Calibration of the Vasicek model

5.7.1 Free calibration on the AAA spot curve

To evaluate how well Vasicek can reproduce an observed term structure, we calibrate model parameters to a real yield curve. The data used are euro area AAA government zero-coupon spot rates published daily by the European Central Bank (ECB).

Data are retrieved automatically via the ECB SDMX API. In practice, some maturities may be updated on slightly different dates (calendar effects, holidays, series availability). To obtain a consistent curve, we keep only points corresponding to the *most frequent date* among the retrieved series. This date (market snapshot) is displayed in the notebook and used as the reference for all calibration figures. A CSV snapshot can also be saved to freeze the exact data used (auditability/reproducibility).

Calibration is performed from a single observed curve on a given date, minimizing a

weighted least squares criterion between model and market spot rates:

$$\min_{(a,b,\sigma,r_0)} \sum_i w_i \left(R^{\text{mod}}(0, T_i) - R^{\text{mkt}}(0, T_i) \right)^2,$$

where weights are chosen proportional to $1/T_i$ to emphasize short maturities.

Figure 8 compares the market spot curve to the Vasicek curve after calibration.

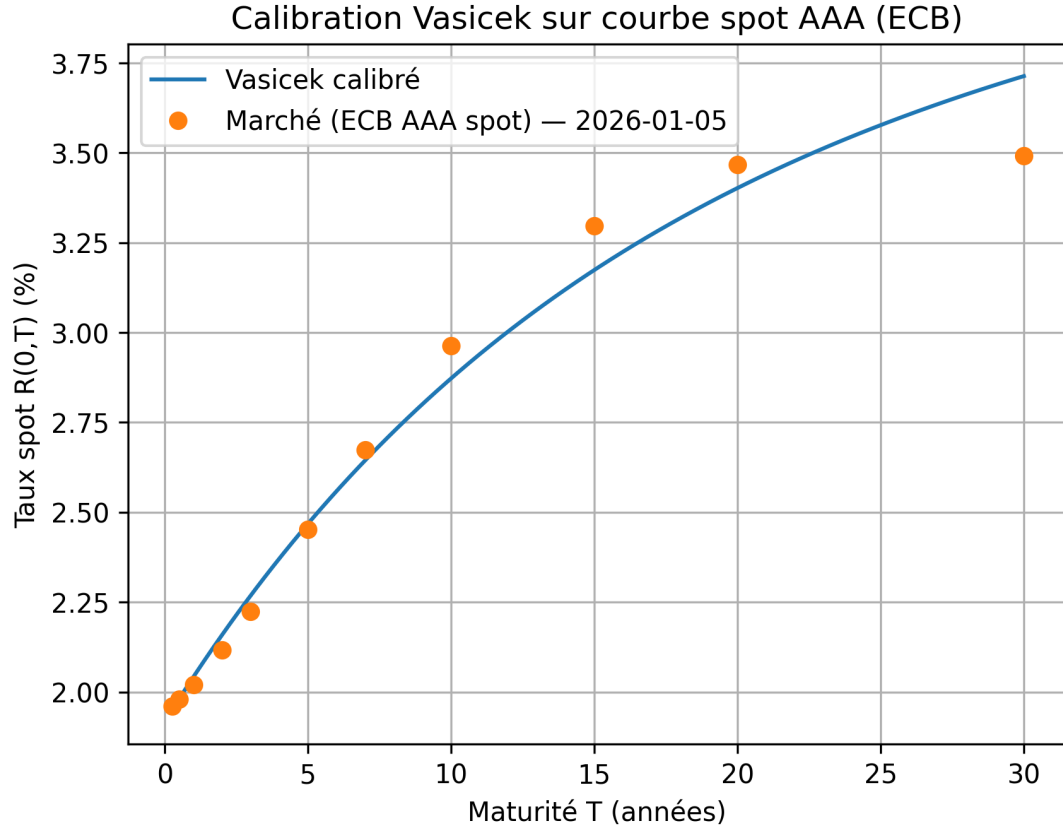


Figure 8: Free calibration of the Vasicek model to the AAA spot curve (ECB).

Figure 9 shows calibration residuals in basis points.

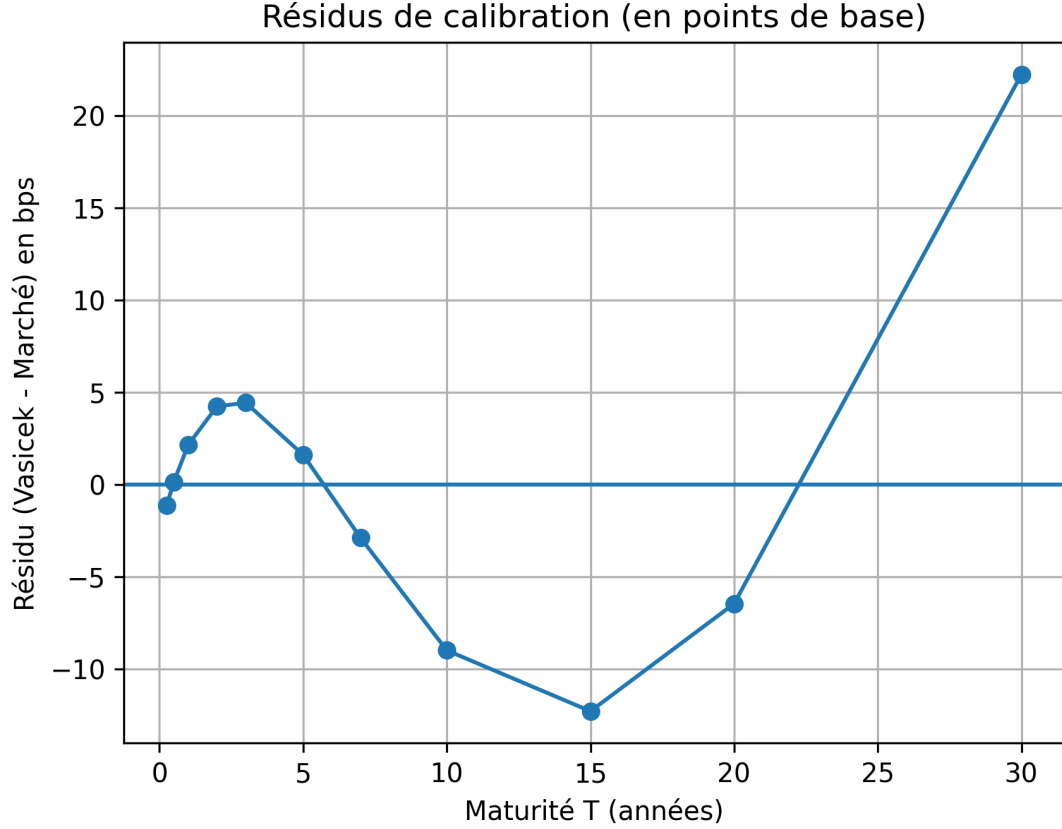


Figure 9: Calibration residuals (Vasicek – market) in basis points.

The calibration yields approximately:

$$a \approx 0.096, \quad b \approx 4.59\%, \quad \sigma \approx 0, \quad r_0 \approx 1.92\%.$$

We observe the model reproduces the overall shape of the spot curve well, with small residuals across maturities. However, the calibrated volatility σ is numerically near zero.

This arises because calibration is performed on a single spot curve at one date. In Vasicek, σ influences the term structure only through a smooth deterministic term, making it difficult to identify from a single spot curve.

As a result, the free calibration tends to favor an almost deterministic short-rate dynamics: parameters a and b are enough to explain the observed curve shape. This highlights a structural limitation of calibrating Vasicek to a single yield curve.

5.7.2 Calibration with fixed volatility

Free calibration on a single spot curve often yields a near-zero volatility, reflecting weak identification of σ from a single term structure. In Vasicek, volatility affects the curve via

a smooth deterministic term, making estimation unstable when only zero-coupon rates are available.

To obtain a more economically realistic calibration, we consider an alternative approach where volatility is *fixed* to a plausible value, and we calibrate only (a, b, r_0) :

$$\min_{(a,b,r_0)} \sum_i w_i \left(R^{\text{mod}}(0, T_i) - R^{\text{mkt}}(0, T_i) \right)^2, \quad \sigma \text{ fixed.}$$

Figure 10 shows the fit with fixed σ , and Figure 11 shows residuals in basis points.

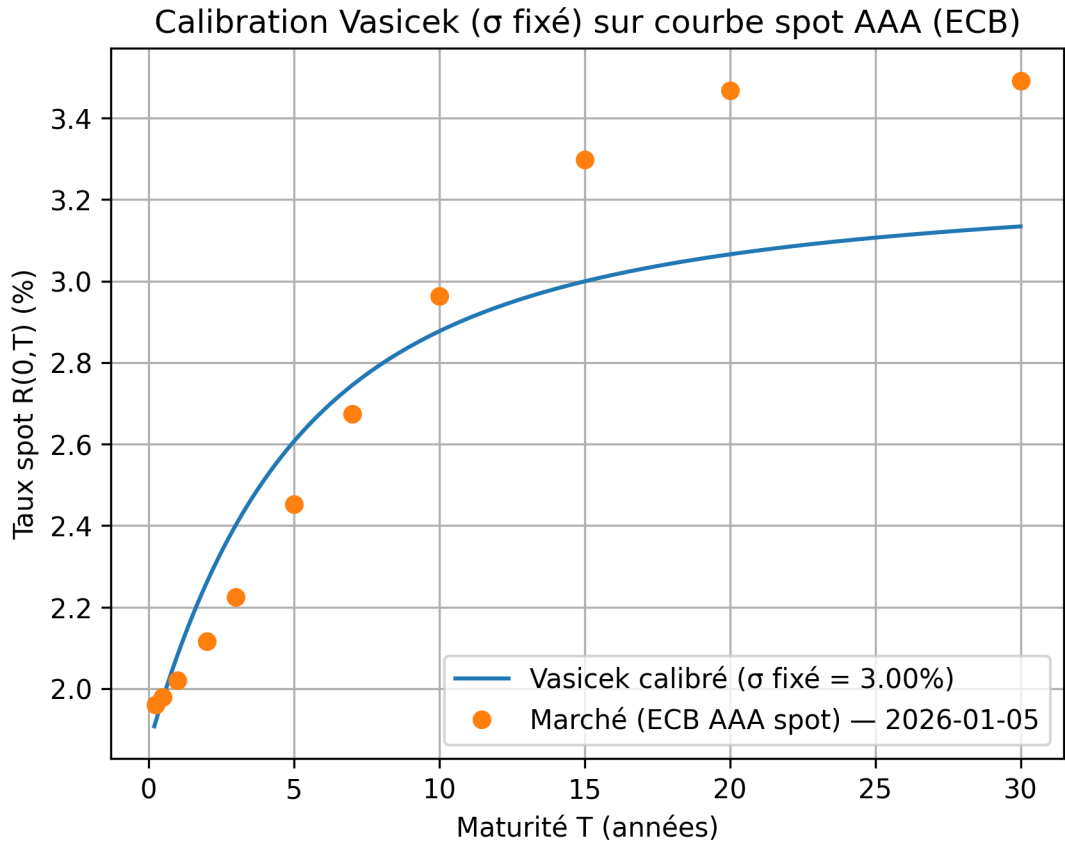


Figure 10: Calibration of Vasicek to the AAA spot curve (ECB), with fixed σ .

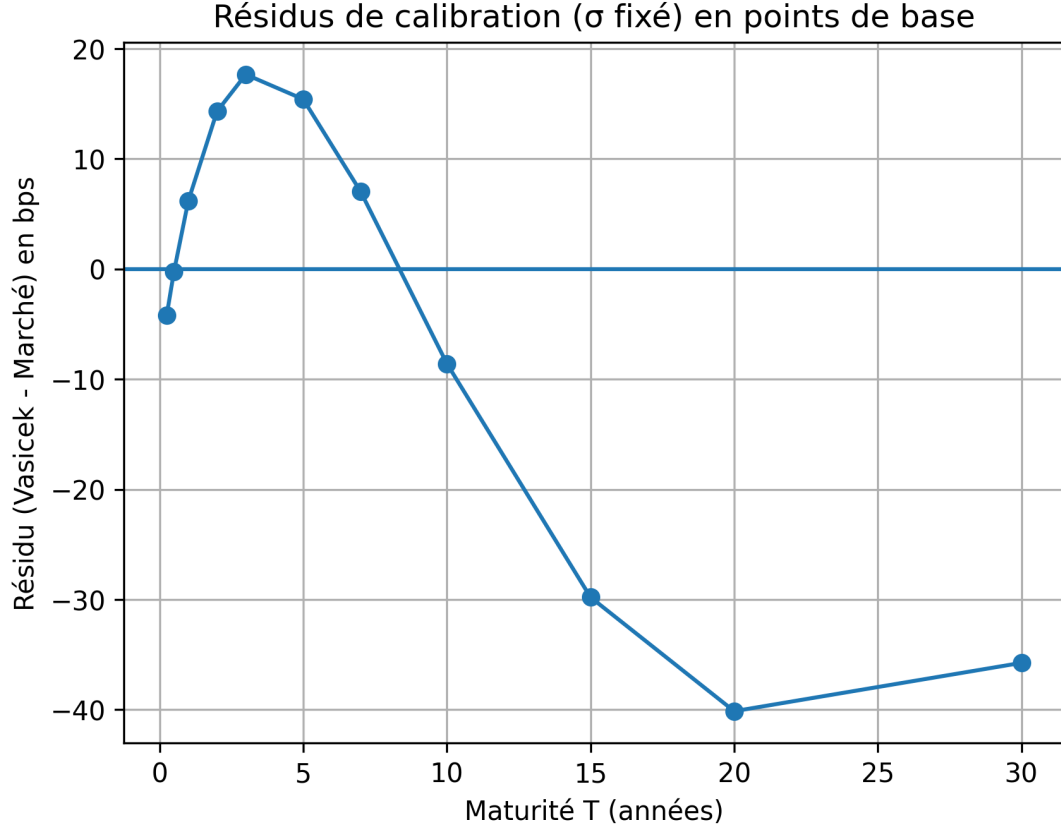


Figure 11: Calibration residuals (Vasicek – market) in basis points, with fixed σ .

The spot-curve fit remains broadly satisfactory while avoiding a degenerate volatility. This is consistent with market practice: σ is typically calibrated to option instruments (caps/floors, swaptions) or to a time series of short rates, while the spot curve mainly pins down level and shape (via a , b , and r_0).

5.7.3 Synthetic comparison: free calibration vs fixed σ

Both calibrations aim to fit the spot term structure, but differ in how much freedom is given to volatility.

Overall fit quality. A simple quantitative metric is the RMSE:

$$\text{RMSE} = \left(\frac{1}{n} \sum_{i=1}^n (R^{\text{mod}}(0, T_i) - R^{\text{mkt}}(0, T_i))^2 \right)^{1/2}.$$

In the notebook, this is reported in basis points.

Residual structure. Beyond RMSE, the residual pattern versus maturity reveals whether error is concentrated in the short end, the long end, or reflects a failure to reproduce specific curvature.

Interpretation. In practice, free calibration often yields very small (near-zero) σ when only a spot curve is used. Fixing σ avoids degeneracy at the cost of slightly reduced flexibility. This aligns with market practice: volatility is identified from options or time-series data, while the spot curve mainly fits level and shape through a , b , and r_0 .

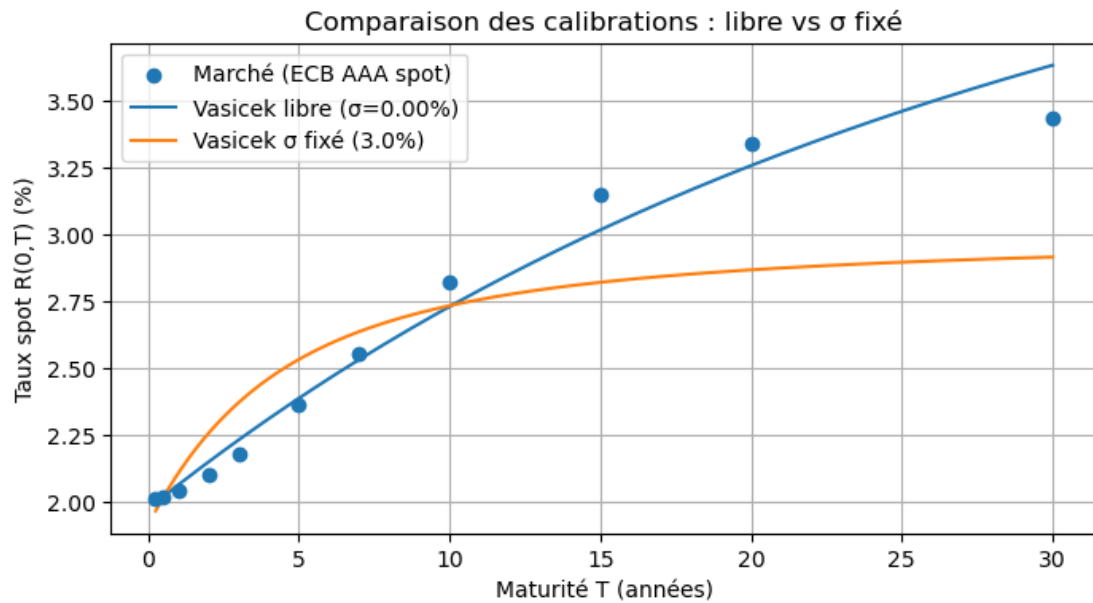


Figure 12: Comparison of fitted curves: free calibration vs calibration with fixed σ .

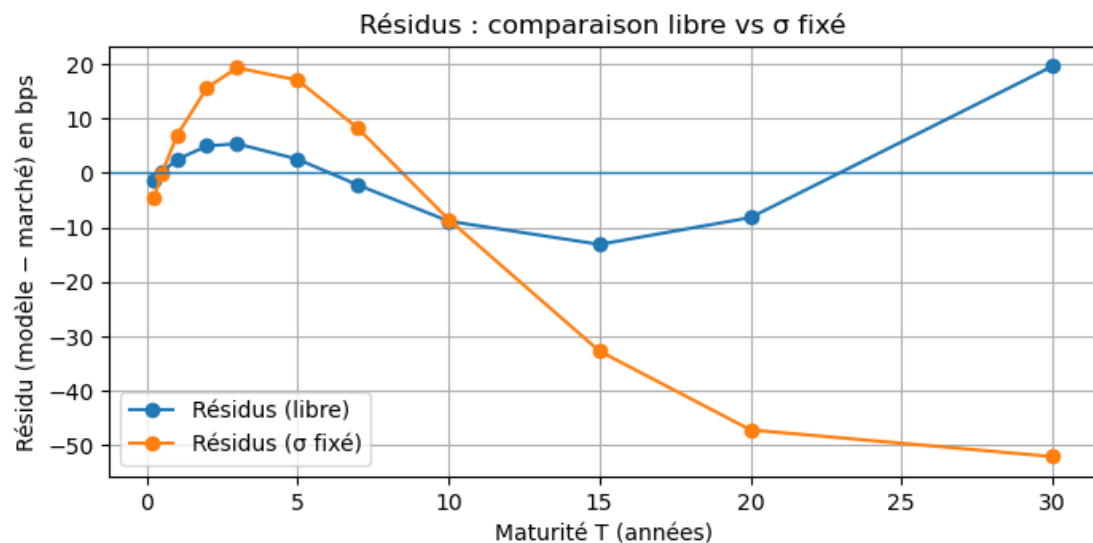


Figure 13: Comparison of residuals (bps): free calibration vs fixed σ .

Practical remark. The two figures above correspond directly to the comparison plots produced in the notebook (calibration section) and visualize the trade-off: more flexibility (free) vs better robustness/interpretability (fixed σ).

6 Quantitative parameter analysis

6.1 Impact of the long-term level b

The parameter b represents the long-term level toward which the short rate reverts. It plays a central role in determining long-maturity yields.

Figure 14 illustrates the effect of b on the spot curve $R(0, T)$ for different b values, with other parameters held constant.

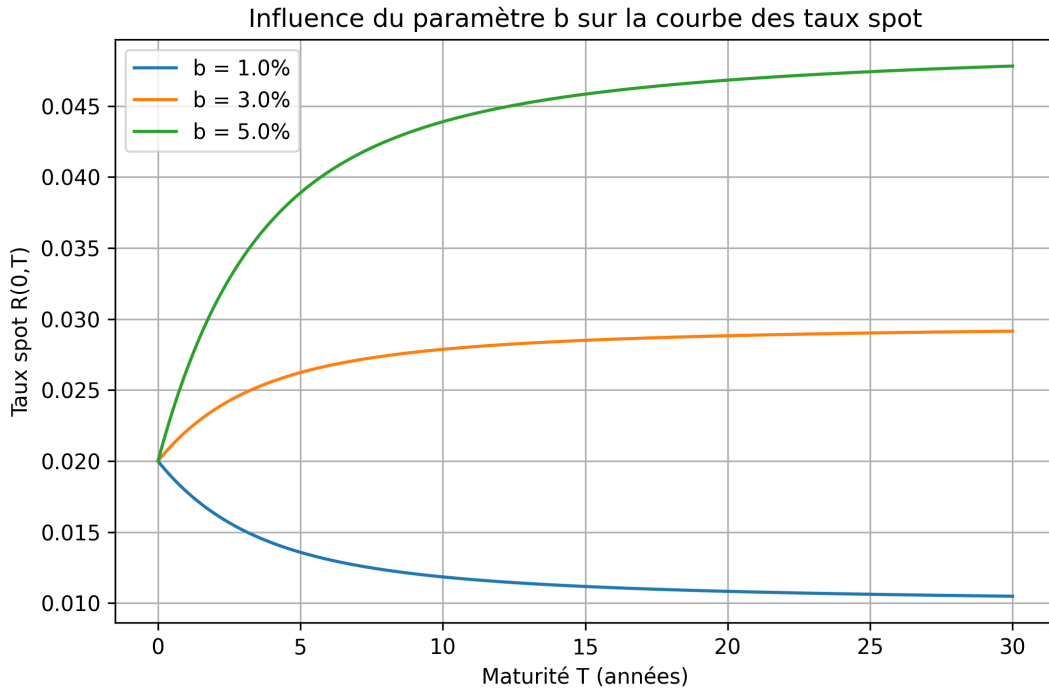


Figure 14: Impact of parameter b on the spot curve $R(0, T)$ in the Vasicek model.

Increasing b produces an almost uniform upward shift of long-maturity yields. The influence of the initial short rate r_0 fades with maturity, and long rates converge to a level primarily driven by b .

6.2 Impact of the mean-reversion speed a

The parameter a controls the speed at which the short rate reverts to its long-term level b . It determines shock persistence and how strongly r_0 influences the term structure.

Figure 15 shows the effect of different a values on the spot curve $R(0, T)$, holding other parameters fixed.

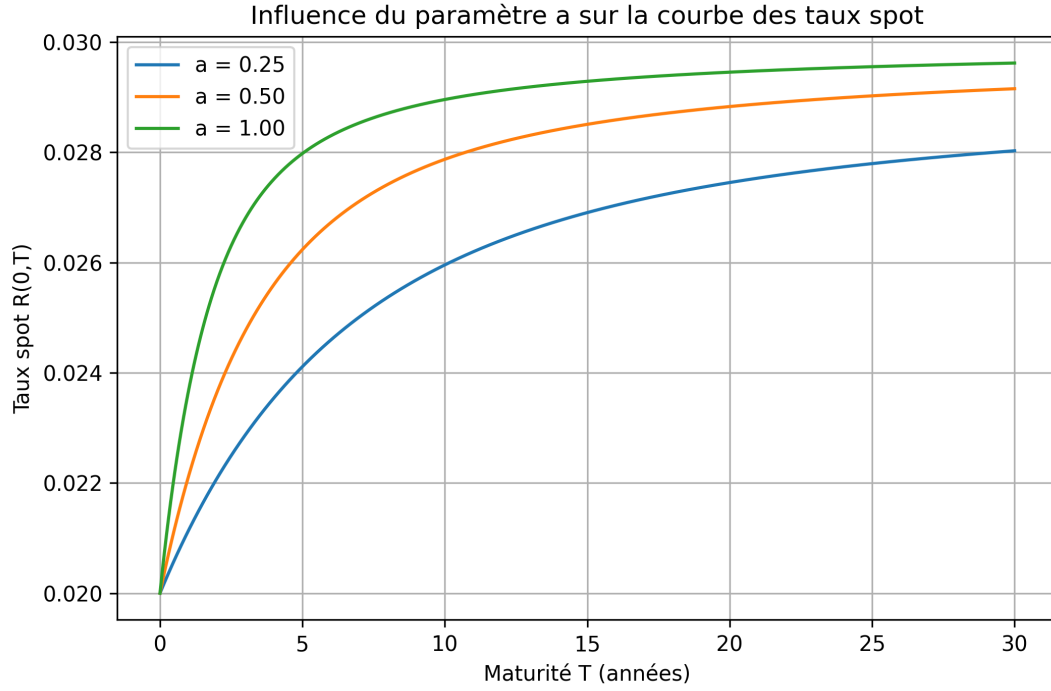


Figure 15: Impact of parameter a on the spot curve $R(0, T)$ in the Vasicek model.

When a is small, mean reversion is slow and the initial short rate influences yields over longer maturities. When a is large, memory of r_0 fades quickly and the curve converges faster to a behavior dominated by b .

6.3 Impact of volatility σ

The parameter σ measures short-rate volatility, i.e. the amplitude of random fluctuations around the mean path. Unlike a and b , volatility does not directly affect the mean dynamics, but it influences bond pricing through uncertainty about future rates.

Figure 16 illustrates the impact of different σ values on the spot curve $R(0, T)$.

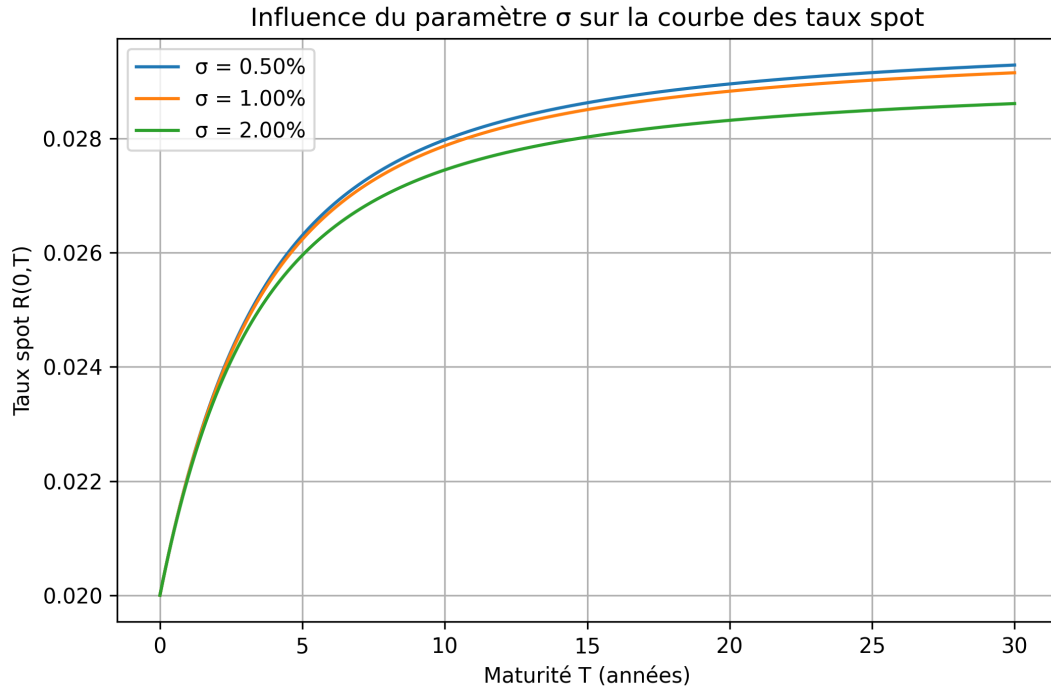


Figure 16: Impact of parameter σ on the spot curve $R(0, T)$ in the Vasicek model.

Higher volatility modifies the term structure, especially at intermediate and long maturities. Increased uncertainty lowers bond prices and thus raises the corresponding yields. This highlights the convexity effect in Vasicek and the role of volatility in shaping the curve beyond mean dynamics.

7 Discussion / model limitations

7.1 Negative rates

A well-known limitation of Vasicek is that the short rate is Gaussian, allowing negative values with non-zero probability. Although long viewed as a major drawback, it became less problematic in the low/negative rate environment observed in some markets since the 2010s.

Still, the possibility of very negative rates may lead to unrealistic behavior at long maturities or over long horizons. This motivates alternative models that enforce positivity, such as the Cox–Ingersoll–Ross (CIR) model.

7.2 Constant volatility

Vasicek assumes constant short-rate volatility, independent of the rate level. This simplifies analysis but is restrictive in practice.

Empirically, rate volatility often depends on the level of rates, typically increasing when rates are higher. The additive noise in Vasicek cannot capture this effect, limiting its ability to reproduce market dynamics.

7.3 Calibration capacity

Vasicek has few parameters (a, b, σ) , making it easy to estimate and interpret. However, this low-dimensional structure limits its ability to fit an observed yield curve precisely at a given date.

In particular, the model cannot exactly match an arbitrary initial term structure. This is problematic for pricing and risk management applications that require tight calibration to market data.

7.4 Alternative models

Several extensions of Vasicek address some limitations while retaining tractability.

The Cox–Ingersoll–Ross (CIR) model uses volatility proportional to the square root of the rate, ensuring positivity. It remains affine and admits closed-form zero-coupon prices.

The Hull–White model can be seen as a time-dependent extension of Vasicek, where

the long-term level becomes a deterministic function of time. This allows exact calibration to the initial yield curve and is widely used in practice for interest-rate derivatives pricing.

These models illustrate the classic trade-off between analytical simplicity and economic realism in interest-rate modeling.

8 Conclusion

8.1 Key learnings

This project provided a concrete exploration of the link between stochastic modeling, PDEs, and numerical methods. The full chain from short-rate dynamics to asset valuation was studied coherently, from the Vasicek SDE to the numerical solution of the associated pricing PDE.

Mathematically, the project strengthened understanding of affine short-rate models and closed-form bond pricing. Numerically, implementing implicit and Crank–Nicolson schemes enabled a detailed analysis of stability, convergence, and computational cost, validated by comparison to a closed-form benchmark.

Finally, quantitative parameter analysis provided intuition for each parameter’s economic role by directly linking parameter choices to the shape of the term structure.

8.2 Overall relevance of the model

The Vasicek model is a highly pedagogical reference framework for interest-rate modeling. Its analytical simplicity yields closed-form zero-coupon prices, making it an excellent tool to understand fundamental mechanisms behind term-structure formation.

Although restrictive assumptions—especially negative rates and constant volatility—limit direct practical use, Vasicek remains a foundational building block of modern quantitative finance. It naturally leads to more advanced models such as CIR or Hull–White, widely used in trading and risk management.

Thus, beyond its limitations, Vasicek fully plays its role as a “companion model”: offering a simple, coherent, analytically tractable setting connecting mathematical rigor, numerical methods, and financial intuition.

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